ARNAV SHARMA

Computer Science and Engineering

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Computer Science Engineer with experience in **distributed systems, edge, AI, IoT, and SWE**. Proficient in **Python**, with hands-on experience in **C, Go, Grafana, cloud, and CI/CD**. Strong problem-solving background with notable competitive programming achievements. Beyond tech, a **FIDE chess** player and **table tennis** hobbyist. Skilled with team and management skills.

Experience

Software Intern – Edge Servers – Akamai Technologies (Bengaluru, Karnataka) **Jun 2026 – Aug 2026**

- Worked at Edge Servers department, a core component of the CDN infrastructure at the **world's largest CDN** providers, contributed to projects including the **GHost Cache System Repository** serving Akamai's global edge network.
- Developed software on Python and Go to build **low-level production replay systems** for Edge Servers, analyzing cache hits and misses from production logs and replay to detect bugs at edge, improve performance and reduce latency.

Technical Consultant – iMerit Technology Services Pvt. Ltd. (Remote | IN) **Nov 2025 – Jan 2026**

- Freelance consultant collaborating within the global iMerit Scholars network on AI evaluation projects.
- Worked on Orion Project, evaluated multimodal AI image-edit outputs for prompt alignment and logical consistency.

Projects

Production AccessLog Replay Orchestration Tool for CDN *(Python, AWS, K6, JavaScript)* **Aug 2024 - Present**

- **Modular Replay Pipeline:** Architected a 4-component Python pipeline to parse and normalize gigabyte-scale production logs (r-lines & f-lines) while preserving exact temporal ordering and edge-routing behaviors.
- **High-Fidelity Replay Driver:** Developed a k6 load-testing driver executing concurrent Virtual Users to validate HTTP status codes, cache hit/miss paths, and origin turnaround times.
- **Custom Edge Infrastructure:** Replayed traffic against a custom-built CDN hosted on AWS EC2 instances configured as edge servers to benchmark real-world edge routing and caching performance.
  https://github.com/ArnavSharma2908/replay-engine-edge-servers

Mystery Animal Voice Game *(ESP32, Router, INMP441 I2S Microphone, MicroPython)* **Sep 2024 - Oct 2024**

- Created an ESP32-powered Mystery Animal guessing game for Robotics Club@NIE, where user needs to guess the simulated animal by asking limited 20 yes/no questions using an INMP441 I2S microphone and Python backend.
- Designed an end-to-end pipeline where user questions are processed on a python server, evaluated against mystery-animal logic, and a response is returned to drive red/green LED indicators and buzzer-based audio feedback.
- Core game logic and interaction flow were adapted and derived from Google's Mystery Animal experiment (2017).
  https://github.com/googlecreativelab/mystery-animal

Delhi Metro Transit Corporation and ChatBot *(Python, matplotlib, pyttsx, WeatherAPI, SQL)* **Oct 2022 - Jan 2023**

- Developed a Python-based metro guide providing route details, fare estimates, and travel times across about 230 stations using Matplotlib for mapping and MySQL for data storage.
- Incorporated METROBOT, an AI chatbot, for user queries, FAQ, and assistance.
- Designed real-time news, weather information along with in-chat Wikipedia access in Metrobot.
  https://github.com/ArnavSharma2908/Delhi-Metro-Transit-Corporation-and-Chatbot

Education

The National Institute of Engineering, Mysuru **Sep 2023 - Sep 2027**
Bachelor of Engineering in Computer Science and Engineering

Salwan Public School, Mayur Vihar, Delhi **Apr 2021 - Mar 2023**
High School Certificate in Science with CS

Achievements

ICPC Regionalist – 2025 Asia West Amritapuri Multisite Regional Contest **Nov 2025**
Attained selection to the 2025 ICPC Asia West Amritapuri Multisite Regional Contest after clearing the qualifying rounds in one of the most prestigious competitive programming competitions worldwide.

2nd Position at National Competitive Programming Potpourri, Shaastra TechFest at IIT, Madras **Jan 2025**
Competed against the brightest minds nationwide and secured second place in the CP (Competitive Programming) Potpurri event at Shaastra '25 Techfest held at IIT Madras.

1st Position at Brainiacs Track, Codeft 5.0 Hackathon, Army Institute of Technology, Pune **Nov 2025**
Won 1st place at Brainiacs Track one of the 4 tracks at Codeft 5.0 Hackathon, AIT Pune by solving tricky logic problems using creative and unconventional thinking instead of standard coding approaches.

Core Skills

Languages and Development Tools: Python, JAVA, GoLang, C, HTML, CSS, JavaScript, LogQL
Tools: Unix, Version Control System (Git), REST APIs, JIRA, Grafana Loki, K6,
Databases: SQL/NoSQL, MySQL, PostgreSQL, MongoDB
Libraries: Matplotlib, Numpy, Langchain, BeautifulSoup, Pyttsx3, Fast API, IronPython, Node.js, Express.js
DevOps: CI/CD, GitHub Actions, Docker, AWS (EC2, S3, EBS Amplify)

Interests & Activities

Chess: FIDE rating (1411); Delhi Zonal Chess Runner-up; VTU Intercollegiate Chess Winner Team (ROB Division)
Clubs & Volunteering: Computer Society Tech Lead at NIE IEEE Student Branch (NISB); Member at Robotics Club; Event Volunteer, Smart India Hackathon – Hardware Nodal Centre; Co-Lead, Kneights Templar chess community